Decent Coaching Classes

VIII CBSE

Mind Your Math VIII

Time: 1 Hour **Marks: 30**

SECTION A (6 × 1 = 6) Choose the correct option.

1. The expression $5x - 2y + 3xy$ is a
   (A) monomial (B) binomial (C) trinomial (D) constant

2. Which of the following is NOT a polynomial?
   (A) $5x^2 - 3$ (B) $3x + \sqrt{x} + 4$ (C) $x^2 + 2x + 1$ (D) 7

3. The common factor of $6x^2y$ and $9xy^2$ is
   (A) $3xy$ (B) $6xy$ (C) $3x^2y^2$ (D) $18x^2y^2$

4. The completely factorised form of $4x^2 + 12x$ is
   (A) $4(x^2 + 3x)$ (B) $2x(2x + 6)$ (C) $4x(x + 3)$ (D) $x(4x + 12)$

5. The solution of the equation $3x - 5 = 10$ is
   (A) 15 (B) 5/3 (C) 3 (D) 5

6. If $2x/3 = 8$, then $x =$
   (A) 16/3 (B) 12 (C) 24 (D) 11

SECTION B (4 × 2 = 8) Answer the following questions.

7. Classify the following as monomial, binomial or trinomial: (a) $3a$ (b) $2x + y$ (c) $5x - 2y + 3xy$ (d) $a^2 + a - 5$

8. Subtract $4a + 5ab - 7$ from $3a - 2ab - 6$.

9. Factorise: $12x^2y - 18xy^2$

10. Solve: $5x - 7 = 3x + 9$

SECTION C (4 × 3 = 12) Answer the following questions.

11. Find the product: $(3a + 5b)(5a + 7b)$
    OR Find the product: $(4x + y)(7x - 6y)$

12. Simplify: $6x(x - 3) + x(2 - 5x) + 3x^2$

13. Factorise: (i) $a^2 + 10a + 25$ (ii) $16x^2 - 9y^2$ (iii) $3x^2 - 12$

14. Solve and check the result: $2(x + 3) - 5 = 3x - 4$
    OR Solve and check the result: $x/2 + x/3 = 10$

SECTION D (1 × 4 = 4) Case-based question.

15. A rectangular plot in a village has length $(2x + 3)$ metres and breadth $(x + 4)$ metres.
    (i) Write the perimeter of the plot in terms of x. (1)
    (ii) If the perimeter of the plot is 50 m, find the value of x. (1)
    (iii) Find the area of the plot as an expression in x (expand the product). (1)
    (iv) Using the value of x from part (ii), find the length and breadth of the plot. (1)